

Hasan Helal

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Education

University of California, Los Angeles

Los Angeles, CA

Bachelor of Science in Physics

Sept 2025 – Present

- Completed the Transfer Research Entry Program
- Completed upper-division courses in Classical Mechanics, Electricity & Magnetism, Quantum Mechanics, and Complex Analysis
- Completed upper-division labs in Machine Learning and Biophysics

De Anza College

Cupertino, CA

Associate in Science in Mathematics for Transfer

Sept 2023 – May 2025

- GPA: 3.97/4.0
- Completed the core IGETC curriculum
- Full physics series: Kinematics & Mechanics, Electricity & Magnetism, Optics & Thermodynamics, and Modern Physics, with associated lab work
- Full mathematics series through multivariable calculus, linear algebra, and differential equations
- Coursework in Python, C++, and MATLAB, plus the chemistry series with associated lab work

Experience

Undergraduate Researcher

Los Angeles, CA

Schaeffer Research Group, UCLA

Oct 2025 – Present

Computational research on magnetized collisionless shocks

- Developed a working knowledge of the particle-in-cell (PIC) code OSIRIS
- Building a forward model to generate synthetic Thomson-scattering spectra from OSIRIS output
- Developing simulation modules for shocks in hybrid and fully kinetic PIC
- Present research progress in weekly group meetings

REU Researcher

Los Angeles, CA

UCLA Physics REU – Schaeffer Research Group

Jun 2025 – Aug 2025

Software development and cross-code benchmarking for a laser-energy-deposition module

- Implemented and validated a laser-energy-deposition module targeting the PIC and radiation-hydrodynamics codes WarpX, OSIRIS, and FLASH
- Built and ran cross-code verification benchmarks, comparing outputs against analytic and published reference cases
- Work centered on numerical implementation and performance

Technical Assistant

Mountain View, CA

Foothill College

Dec 2024

Helped run the departmental Physics Show

- Explained physics phenomena to elementary-school children in an accessible way
- Led campus tours highlighting community-college opportunities for low-income students

Junior Researcher

Oct 2024 – Dec 2024

Stanford Small Science Groups

Informal mentored research project

- Worked with a senior mentor on a self-directed question about quantum spin
- Presented findings at the conclusion of the program

Projects

Magnetized Collisionless Shocks in WarpX

[GitHub](#) 

- Reproduced and verified the piston-driven magnetized shocks of Schaeffer et al. (2020) in the particle-in-cell code WarpX (KinShock2020)
- Ported a ray-traced laser-deposition module (after Hyder et al. 2024) to drive shocks from a modeled laser rather than a prescribed piston (LaserProdShock)
- Extended the shock-driver operators to WarpX's hybrid kinetic-ion/fluid-electron solver, cutting simulation cost by over three orders of magnitude versus explicit PIC (H-PICShock)

Synthetic Thomson Spectra from PIC Simulations

[GitHub](#) 

- Built a forward-model pipeline that takes output from a general PIC simulation and generates Thomson spectra
- Generalized the pipeline to both OSIRIS and WarpX
- Tools: Python, OSIRIS, WarpX

Machine Learning for Thomson-Scattering Diagnostics

[GitHub](#) 

- Built a dense neural network to infer plasma parameters from Thomson-scattering spectra
- Compared the efficacy of transfer learning against traditional fitting methods
- Tools: Python, PyTorch

Computational Physics Simulations

[GitHub](#) 

- Python and Jupyter simulations across classical mechanics, electromagnetism, fluid dynamics, thermodynamics, and quantum mechanics
- n -body orbital dynamics: specific solutions to the n -body problem and Barnes–Hut galaxy simulation
- Self-consistent atomic and molecular density-functional-theory (Hartree–Fock / Kohn–Sham LDA) solver
- Tools: Python, NumPy, SciPy

General-Relativistic Raytracing

[GitHub](#) 

- Raytracer through Schwarzschild spacetime rendering gravitational lensing, the photon sphere, the black-hole shadow, and a lensed accretion disk
- Extending toward Kerr (rotating) spacetimes and gravitational-wave inspiral modeling
- Tools: Python

OpenGL Physics

- Interactive C++/OpenGL physics simulators sharing a fixed-timestep framework (GLFW, glad, Dear ImGui)
- Live atomic Hartree–Fock / Kohn–Sham LDA SCF solver, N-body gravity, and 3D Lennard-Jones molecular dynamics
- Tools: C++, OpenGL, CMake

Quantum Mechanics Substack

[Substack](#) 

A visual introduction to the Schrödinger equation

- Use simulation and analogy to explain the core equation of quantum mechanics to a lay audience
- Developed the physics simulator behind the quantum-mechanical visualizations

Skills

Physics: Quantum Mechanics, Optics, Classical Mechanics, Classical Electrodynamics, Special Relativity, Statistical Mechanics

Programming: Python, Machine Learning (PyTorch), C++, MATLAB, L^AT_EX, Git, Excel

Mathematics: Differential equations, calculus, linear algebra, complex analysis, and group theory; working familiarity with differential geometry

Languages: English (fluent), Arabic (native)